Shuva Narayan Gautam

343-297-9767 | gautns7@gmail.com | linkedin.com/in/shuvagautam/ | github.com/NarayanGautam

Education

Carleton University — Ottawa, Ontario, Canada
Bachelor of Computer Science, Co-op, Artificial Intelligence/Machine Learning Stream (3.94/4.0 GPA) — *Sept. 2021 – April 2026*

Technical Skills

Languages and Frameworks: Python, Java, C/C++, JavaScript/TypeScript, SQL, HTML/CSS, Bash, React, Node.js, Qt, Robot
Tools/Infrastructure: Docker, Kubernetes, AWS, CI/CD, Git, Linux, MongoDB, PostgreSQL, Selenium, WebdriverIO, JMeter
AI/ML/Data: pandas, NumPy, scikit-learn, TensorFlow, OpenCV, YOLO, NetworkX, LLM APIs, RAG, prompt engineering, agentic coding

Work Experience

Network Automation Intern — January 2025 – April 2025
Nokia — *Python, Power Platform, Power BI*

- Maintained the network automation use-case catalogue with Python, Power Platform, and Power BI, streamlining the documentation and accessibility of use cases for internal teams and customers
- Recorded and produced use case demos for Nokia clients, showcasing the practical applications and benefits of network automation solutions

Software Test Specialist Intern — May 2024 – August 2024
Nokia — *Mocha, Selenium, WebdriverIO, OpenStack*

- Automated Network Service Platform testing with Mocha, Selenium, and WebdriverIO across simulated management and network scenarios
- Built network setups to streamline feature development and testing using OpenStack and a virtualized environment for network functions
- Triaged regression failures, improved script reliability, and delivered fixes within an Agile Scrum team

Software Quality Assurance Analyst Intern — May 2023 – December 2023
Environment and Climate Change Canada — *Robot Framework, Selenium, JMeter, LoadRunner, Applitools*

- Expanded test coverage with Robot Framework and Selenium automation, reducing manual testing time
- Performed load and performance testing using JMeter and LoadRunner to ensure the applications can handle the expected load and stress
- Evaluated Applitools for visual AI testing to simplify validation of UI elements across applications

Teaching Assistant — Fall 2025; Fall 2022 – Winter 2023
Carleton University — *Java, Python*

- Led labs and office hours for COMP1805 (Fall 2025), COMP1406 (Java/OOP), and COMP1405 (Python); guided students through coursework and debugging
- Graded assignments and provided constructive feedback on code quality and problem-solving approach

Projects

Traffic Flow Forecasting | *Python, OpenCV, YOLO, LSTM* — April 2026

- Built a computer vision pipeline with YOLO object detection, multi-object tracking, and automated line-cross counting to extract vehicle volume data from traffic camera footage, enabling real-time traffic monitoring
- Developed and compared baseline (ARIMA) and LSTM models to forecast traffic flow, outperforming traditional statistical methods

Autonomous Robot Navigation Simulation | *Java, Webots, LiDAR* — December 2024

- Simulated an autonomous robot in Webots with sensor inputs to navigate a maze environment, locating and collecting jars and delivering them to a drop-off point
- Implemented a pathfinding algorithm to navigate the maze, using LiDAR sensor data to detect and avoid obstacles while efficiently reaching target locations

Neureset EEG Simulator | *Qt, C++, QCustomPlot* — April 2024

- Designed and created a Qt/C++ desktop application for patient management and therapy session scheduling
- Gathered requirements, created use cases and UML class and sequence diagrams to model the application to streamline the development process